



Lecture 9

Fungi and Fungal Diseases

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Intended Learning Outcomes (ILOs)

By the end of this lecture, the student will be able to:

1. Describe the fundamentals of general mycology, including:

- Structural characteristics
- Morphological classification
- Medical significance
- Treatment approaches

2. Discuss *Candida albicans* with its clinical relevance and associated infections.

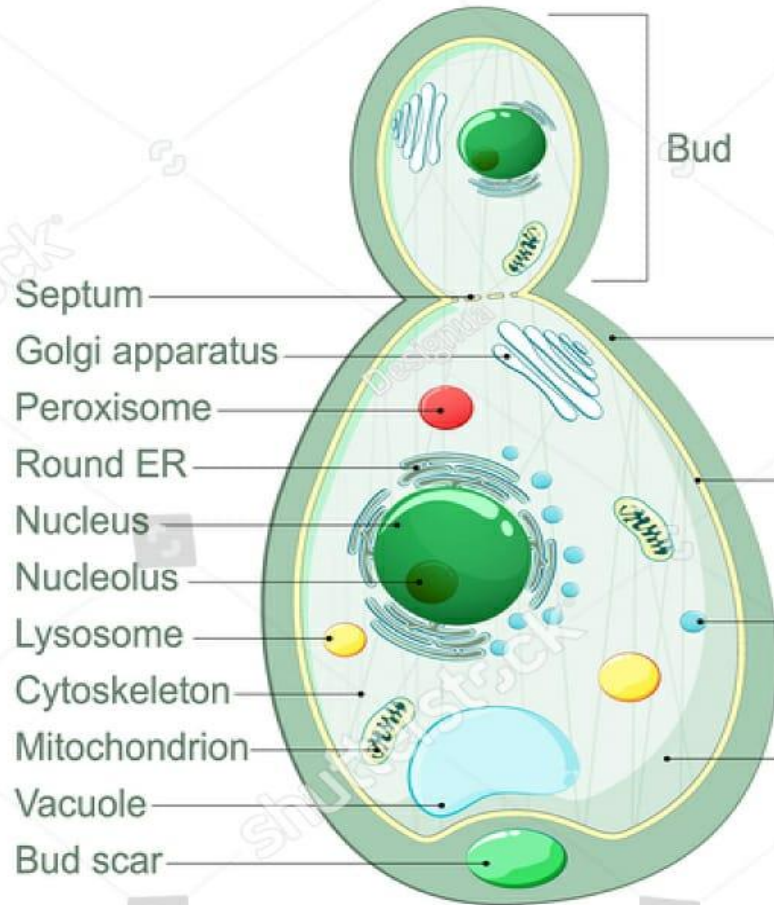
Fungi € eukaryotic, bacteria are € prokaryotic

Character	- Prokaryotes - Bacterial cells	- Eukaryotes - Fungal cells
1. Nucleus	Absent	Present
• Nuclear membrane		
• Chromosome	1 (circular)	> 1 (linear)
2. Cytoplasm Organelles such as mitochondria & ER	No	Yes
• Size of ribosome	70S	80S
3. Cell membrane Sterol	No	Yes (ergosterol, zymosterol)
4. Cell wall	peptidoglycan	Chitin
5. No. of cells	Unicellular	Unicellular or multicellular

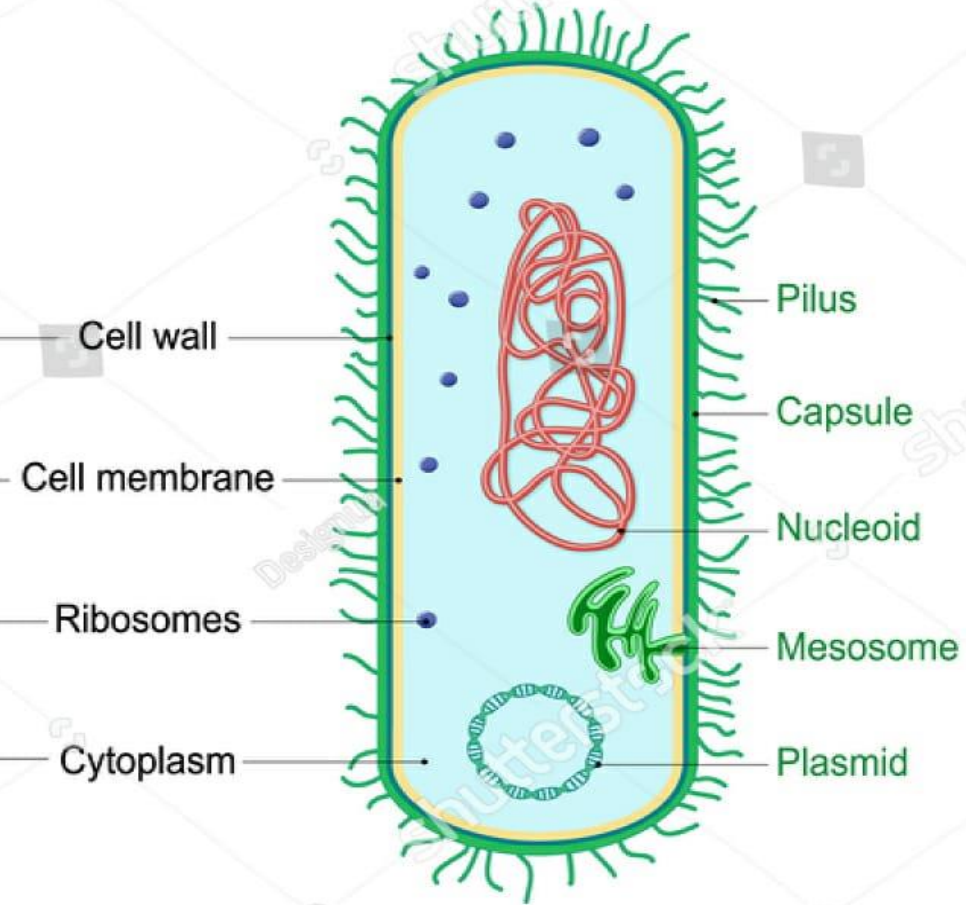
Human cells:

cell membrane contain **cholesterol** – have **no cell wall** – reproduce **sexually** only - **multicellular**

Fungal cell

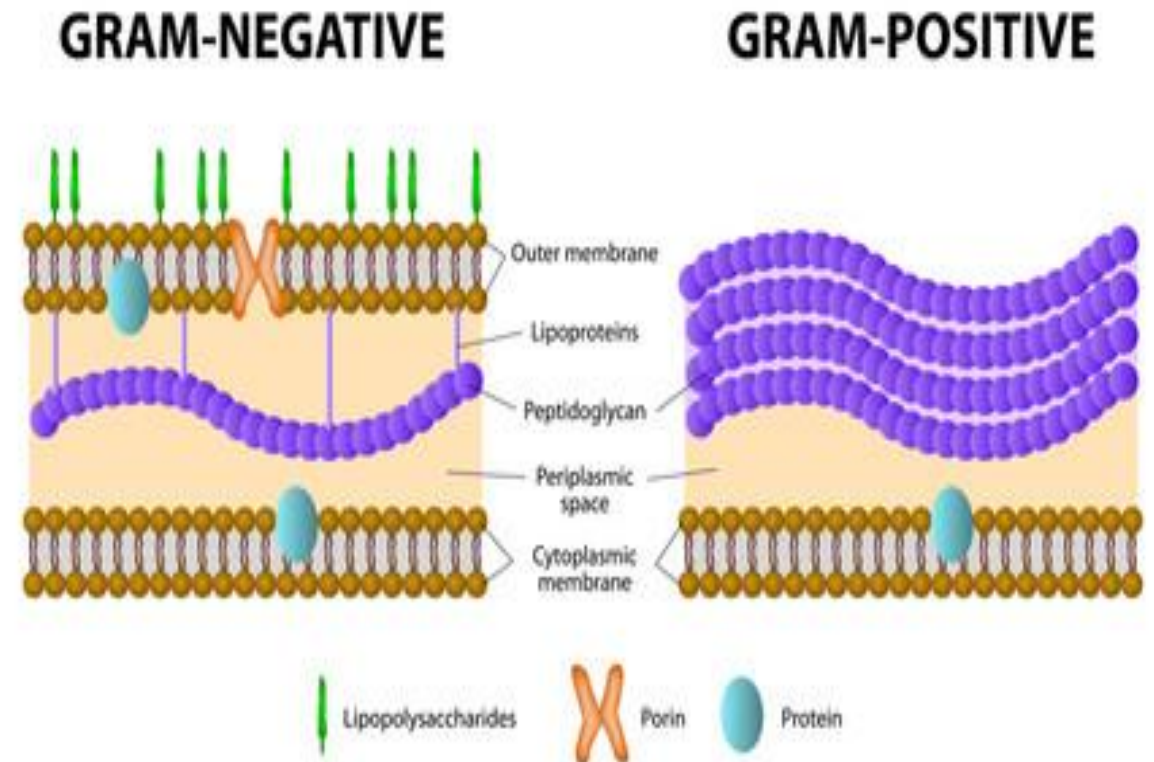
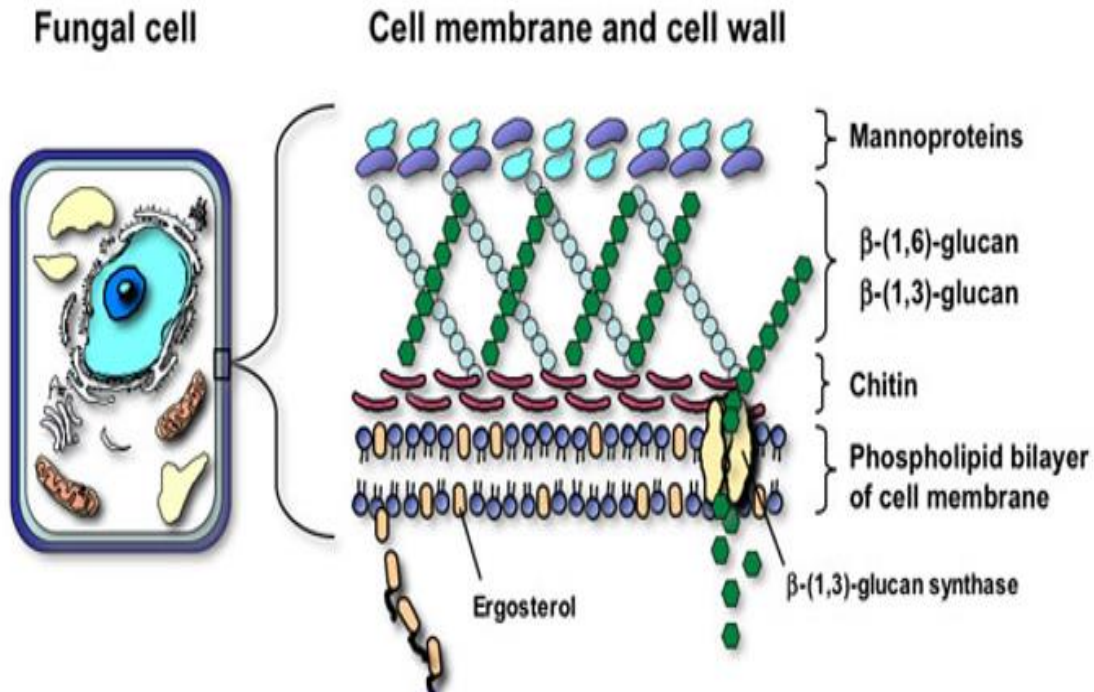


Bacterial cell



Two fungal cell structures are medically important:

	Fungus	Bacteria	Importance
1. Cell wall	Chitin + polysaccharide	Peptidoglycan	Fungi are insensitive to antibiotics



	Fungi	human	Importance
2. Cell membrane	Ergosterols Zymosterol	Cholesterol	Selective action of antifungal drugs acting on mm e.g Fluconazole

Fungus are either unicellular or multicellular. The majority of fungi are multicellular organisms (composed of many cells).



MICROBIOLOGY CHALLENGE

Fungi Basics Mastery 🧪 ✨

- I **Q1** Fungi can exist as:
- A** ONLY UNICELLULAR
 - B** ONLY MULTICELLULAR
 - C** UNICELLULAR OR MULTICELLULAR
 - D** NON-CELLULAR
- I **Q2.** Human cell membranes contain:
- A** ERGOSTEROL
 - B** PEPTIDOGLYCAN
 - C** CHOLESTEROL
 - D** CHITIN



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MICROBIOLOGY CHALLENGE

Antibiotics and Fungi 🧪 ✨

Q1. A nurse is caring for a patient receiving long-term antibiotics who develops oral white patches. What is the most likely cause:

- A** Bacterial infection
- B** Viral infection
- C** Fungal overgrowth
- D** Autoimmune disease

Q2. Which difference explains why antibiotics do not affect fungal cells:

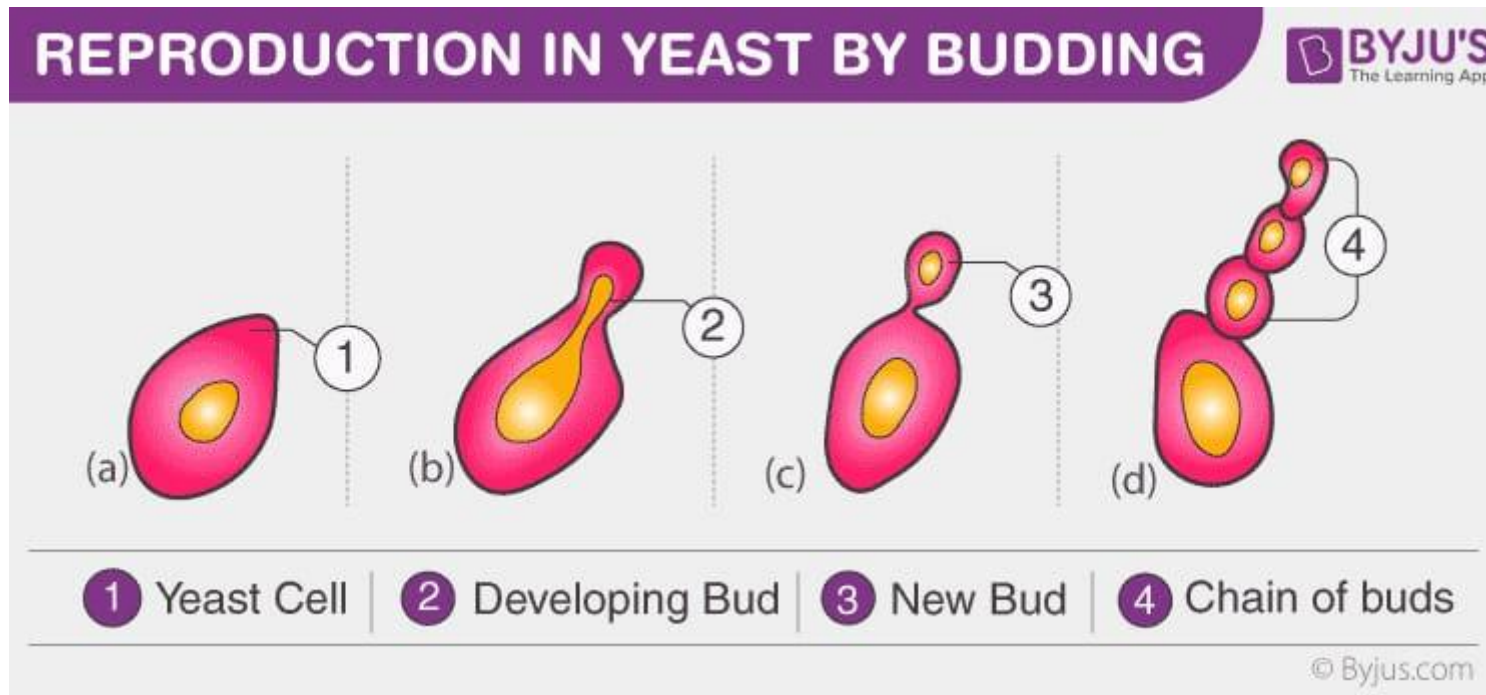
- A** Fungi lack ribosomes
- B** Fungi have different cell wall composition (chitin)
- C** Fungi have circular DNA
- D** Fungi lack cytoplasm



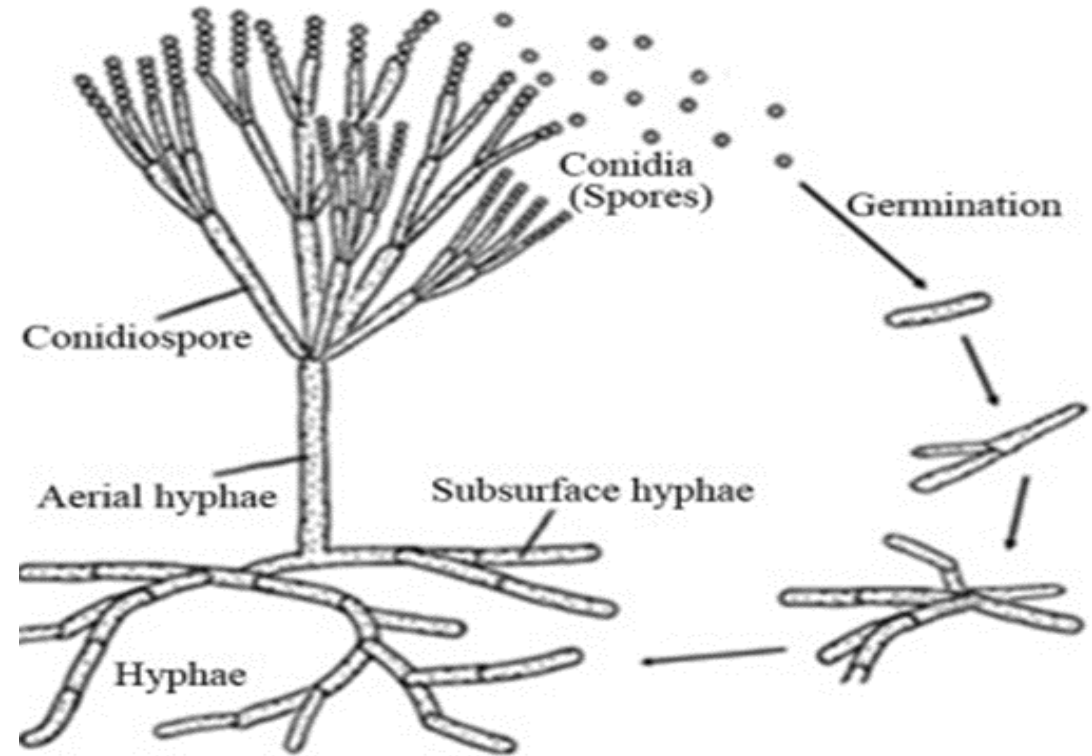
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Fungi,..... Morphological classification

1-Yeast: oval or round, single cells
(unicellular) reproduce asexual by
budding as **candida**



2-Hyphae: tubular
projections of multicellular
fungi that form **mycelium**
(filamentous **network of
hyphae**) and reproduce
asexual by spore as
aspargillus



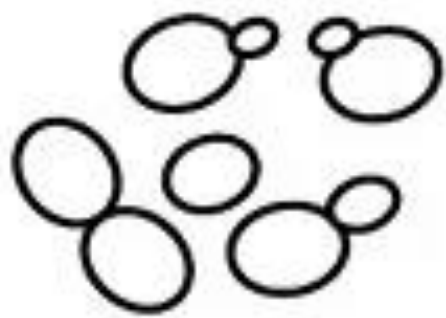
Fungi,..... Morphological classification

3- Dimorphic fungi: is the condition where a fungus can exhibit either **the yeast** form or **the hyphal** form, depending on growth conditions:

-**yeast form:** in tissue or when grown at **37°C**

-**hyphae form:** when grown at **22°C**

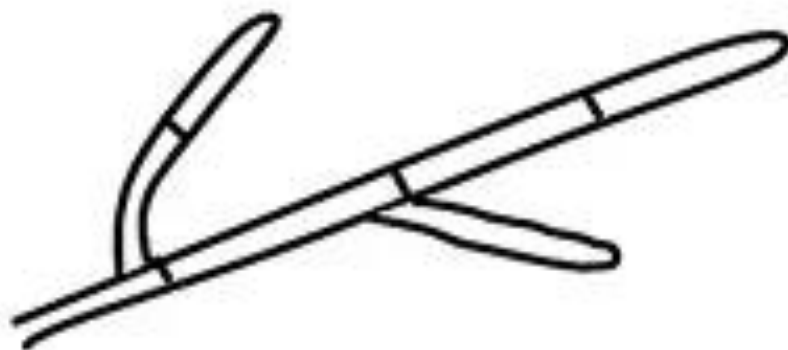




Yeast



Pseudohyphae



Hyphae

Complete

1. The difference between the cell wall of fungi and bacteria:

a- ,b-

2. The three morphological forms of fungi:

a- ,b- ,c-

Fungi,..... Medical importance

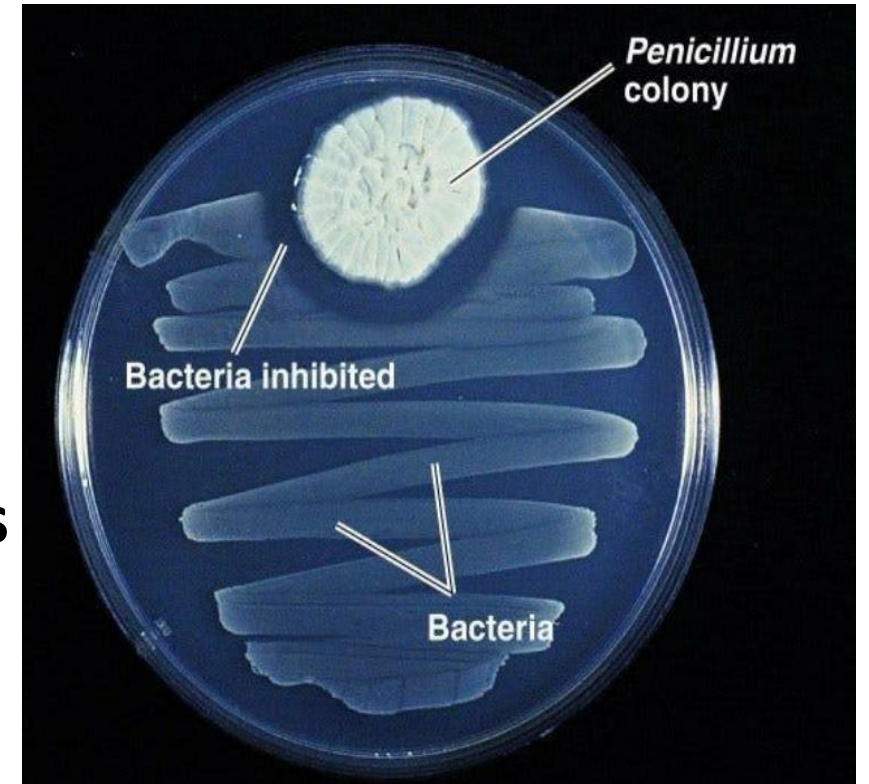
1-Beneficial Effects (Positive Role)

Production of antibiotics

- Penicillium → Penicillin

Production of useful products

- Vitamins, enzymes
- Used in biotechnology & pharmaceuticals



2-Harmful Effects (Pathogenic role)

A- Production of toxin (Mycotoxicosis):

Aflatoxin produced by *Aspergillus flavus* which grows on spoiled grains & peanuts causes hepatocellular carcinoma



B- Fungal infections (mycoses)

1- Superficial:

Mucous membrane in **Oral cavity**

Superficial mycoses involve the **outer layers of the body**, especially keratin-rich tissues
(skin, hair, nails), and are mainly caused by **dermatophytes**.



2- Subcutaneous:

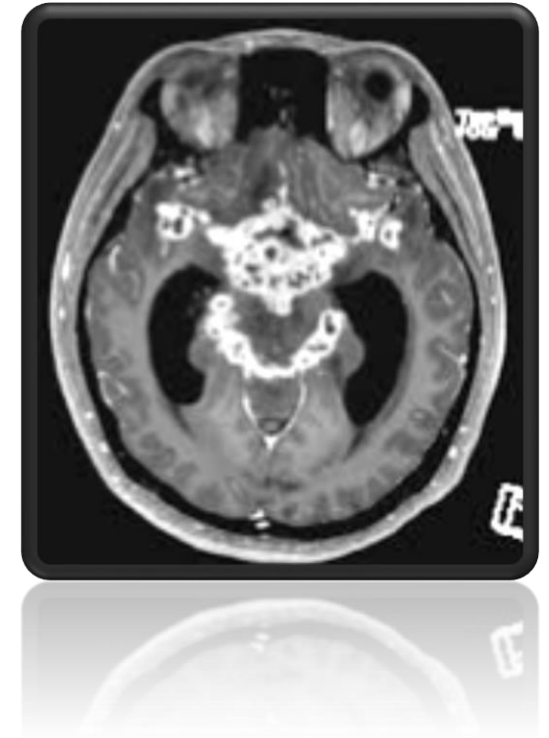
- Traumatic implantation of environmental fungi
- Swelling+ tissue destruction + sinus formation
- Common in tropics as **mycetoma**



3. Systemic:

- Fungi spread through the **bloodstream (hematogenous spread)**
- Affect vital organs: **brain, lungs, blood**
- May cause severe conditions such as **meningitis, fungemia, and lung infection**
- Life-threatening if not treated
- Common in immunocompromised patients

Systemic fungal infections are serious, rapidly spreading, and require early diagnosis and treatment.





MICROBIOLOGY CHALLENGE

Mycoses and Toxins  

I Q1 Aflatoxin is associated with which condition?

- A** LUNG FIBROSIS
- B** HEPATOCELLULAR CARCINOMA
- C** RENAL FAILURE
- D** DIABETES MELLITUS

I Q2 A patient presents with fungal infection affecting hair, nails, and skin. This is classified as:

- A** SYSTEMIC MYCOSIS
- B** SUBCUTANEOUS MYCOSIS
- C** SUPERFICIAL MYCOSIS
- D** OPPORTUNISTIC INFECTION

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MICROBIOLOGY CHALLENGE

Fungal Diagnosis and Transmission 

I Q1 A farmer develops a chronic foot swelling after injury from a thorn. The lesion shows sinus formation. The most likely diagnosis is:

- A** ORAL CANDIDIASIS
- B** MYCETOMA
- C** ASPERGILLOSIS
- D** VIRAL INFECTION

I Q2 Systemic fungal infections spread mainly through:

- A** LYMPHATIC SYSTEM ONLY
- B** SKIN CONTACT
- C** BLOODSTREAM (HEMATOGENOUS SPREAD)
- D** AIR ONLY

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Fungi,..... Chemotherapy

Systemic Antifungal Drugs

1-Amphotericin B

- Broad-spectrum antifungal agent
- Reserved for severe systemic infections
- Use is limited due to nephrotoxicity

2-Azoles (inhibit ergosterol synthesis)

- ❑ Fluconazole
- ❑ Ketoconazole

Topical (Local) Antifungal Drugs

1-Nystatin

Used for superficial fungal infections

Mainly applied to skin and mucous membranes

2-Imidazoles

- ❑ Clotrimazole
- ❑ Miconazole

Commonly used for cutaneous infections

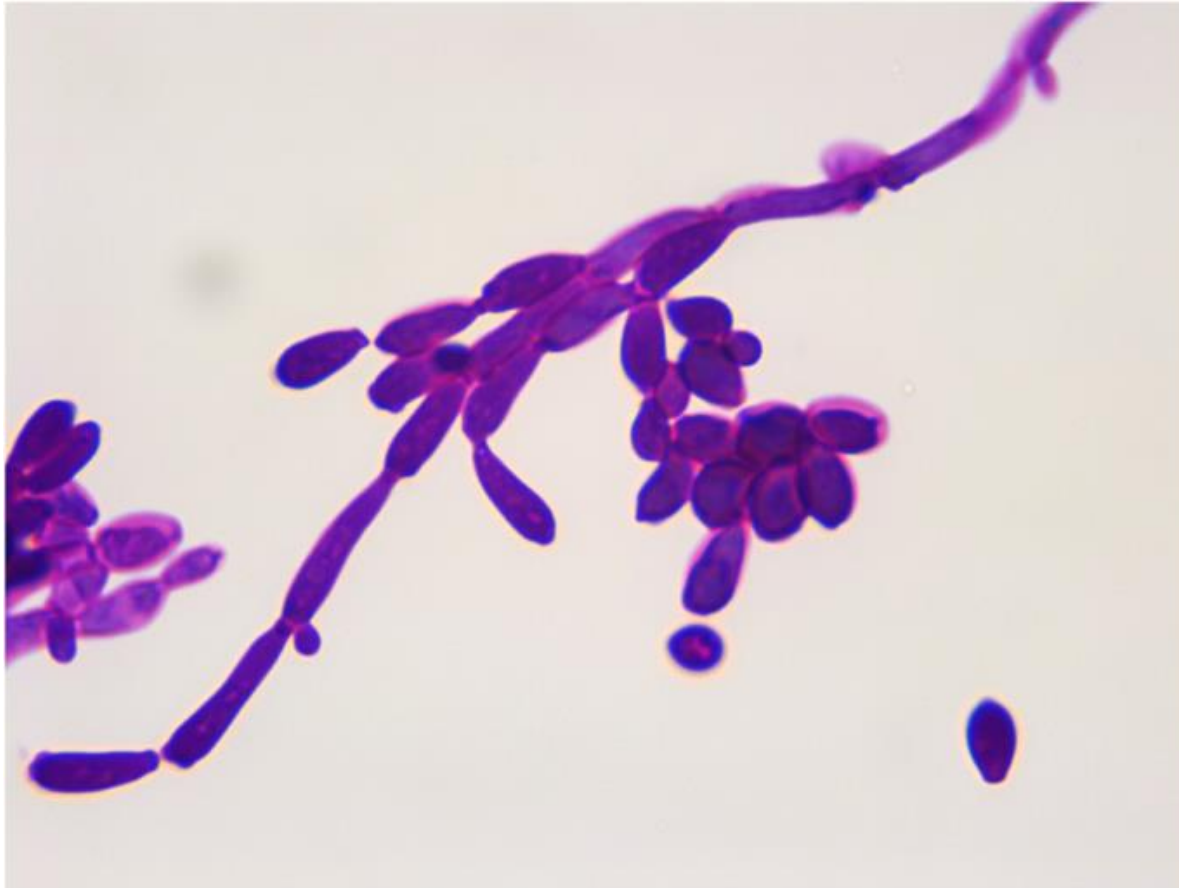


Candida

- Several Candida species colonize humans.
- **Candida albicans** is the most common pathogenic species.
- It is responsible for approximately 90% of Candida infections.



Candida albicans,....Habitat



It is part of the normal flora of mucous membranes of the upper respiratory, gastrointestinal, and female genital tracts.

Candida albicans,....Pathogenesis and clinical findings

1-Source of Infection

Primarily endogenous:

Candida albicans is part of the normal flora of:

- Skin
- Oral cavity
- Oropharynx
- Gastrointestinal tract
- Female genital tract

Infection occurs when there is **disruption** of normal flora or **impaired host immunity**, leading to **overgrowth**.

Exogenous (Cross infection) may occur:

- From mother to neonate during delivery
- Between infants in nurseries



Candida albicans,....Pathogenesis and clinical findings

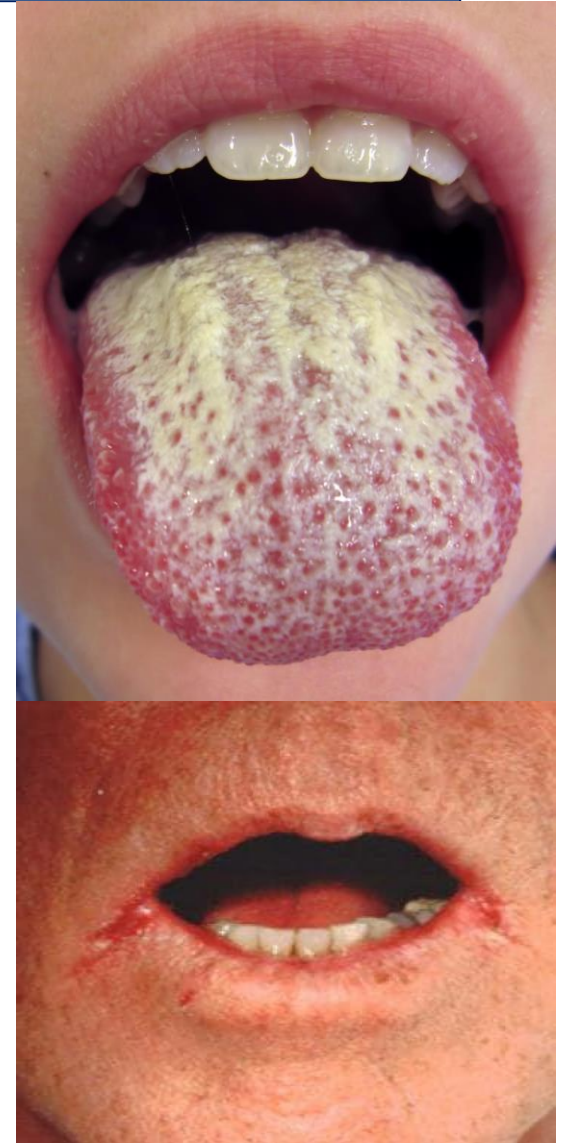
2. Host Factors

- **Neonates**, especially premature infants, are highly susceptible
- **Adults** with predisposing conditions, including:
 - Local factors: use of dentures
 - General conditions: poor health status, particularly diabetes mellitus
- High-risk (**immunocompromised**) patients, such as those with:
 - Diseases: e.g., **AIDS, malignancies, leukemia, lymphoma**
 - Therapies: prolonged use of broad-spectrum antibiotics, immunosuppressive agents, or cytotoxic drugs

Candida albicans,....Pathogenesis and clinical findings

3. Lesions “candidiasis

- **Superficial : skin and mm**
- **Mucous membranes:** at the site where it is naturally present as mouth, throat, gut, and vagina
 - Oral candidiasis:
 - o Called oral thrush, white adherent patches on buccal mucosa; **inner cheeks, tongue, roof of the mouth, and throat**
 - o Candida-associated oral lesions: Angular cheilitis and denture stomatitis



- Superficial:

Skin: red weeping areas

usually when skin is moist

and traumatized e.g

intertrigo in the obese (skin

fold infections)





MICROBIOLOGY CHALLENGE

Candida Risk and Transmission 

I Q1 Which of the following conditions increases susceptibility to candidiasis?

- A** HYPERTENSION
- B** IMMUNOSUPPRESSION
- C** HYPERTHYROIDISM
- D** ASTHMA

I Q2 Cross infection of Candida can occur in hospitals through:

- A** AIR ONLY
- B** FOOD CONTAMINATION
- C** MOTHER TO NEONATE DURING DELIVERY
- D** INSECT BITES

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CASE 1

A nurse observes white, adherent patches on the tongue and inner cheeks of a patient receiving long-term antibiotics.

I Q1 What is the most likely diagnosis?

- A** LEUKOPLAKIA
- B** ORAL CANDIDIASIS
- C** APHTHOUS ULCER
- D** HERPES SIMPLEX

I Q2 What is the main predisposing factor?

- A** VIRAL INFECTION
- B** ANTIBIOTIC THERAPY
- C** TRAUMA
- D** BLOODSTREAM (HEMATOGENOUS SPREAD)



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**Thank you for your
attention!**
